

10(a)	1 ref. to organisms change over time ; <i>plus any six from:</i> 2 organisms produce many offspring ; 3 more than is necessary to maintain population ; 4 (but) population size is constant (over time) ; 5 within a species there is variation (in phenotype) ; 6 due to genetic variation ; 7 caused by mutation ; 8 individuals compete for survival / survival of the fittest ; 9 ref. to selection pressure(s) ; 10 some individuals, are better adapted to survive / have advantageous alleles / have selective advantage ; 11 pass on (advantageous) alleles to offspring ; 12 changes allele frequency ; 13 ref. to speciation ;	7
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Question	Answer	Marks
10(b)	<i>any eight from:</i> <i>meiosis (max seven):</i> 1 chiasma / crossing over ; 2 between <u>non</u>-sister <u>chromatids</u> ; 3 of, homologous chromosomes / bivalent ; 4 in prophase 1 ; 5 exchange of, genetic material / DNA ; 6 linkage groups broken ; 7 new combination of alleles ; 8 random / independent, assortment of, homologous chromosomes / bivalents (at equator) ; 9 (during) metaphase 1 ; 10 random / independent, assortment (of, sister chromatids / chromosomes) at metaphase 2 ; 11 possible (chromosome) mutation ; <i>fertilisation:</i> 12 random mating ; 13 random, fusion / fertilisation, of gametes ;	8